

BACKGROUND

Motivations

- **Early identification of deterioration** is critical for timely intervention and improved patient outcomes.
- Existing severity scores (e.g., NEWS2) rely **primarily on structured data** and may overlook valuable information in clinical notes [1].

Challenges

- Clinical notes are **unstructured** and difficult to incorporate into traditional predictive models.
- Clinical deterioration is inherently **time-dependent**, making binary classification insufficient for capturing temporal risk dynamics.

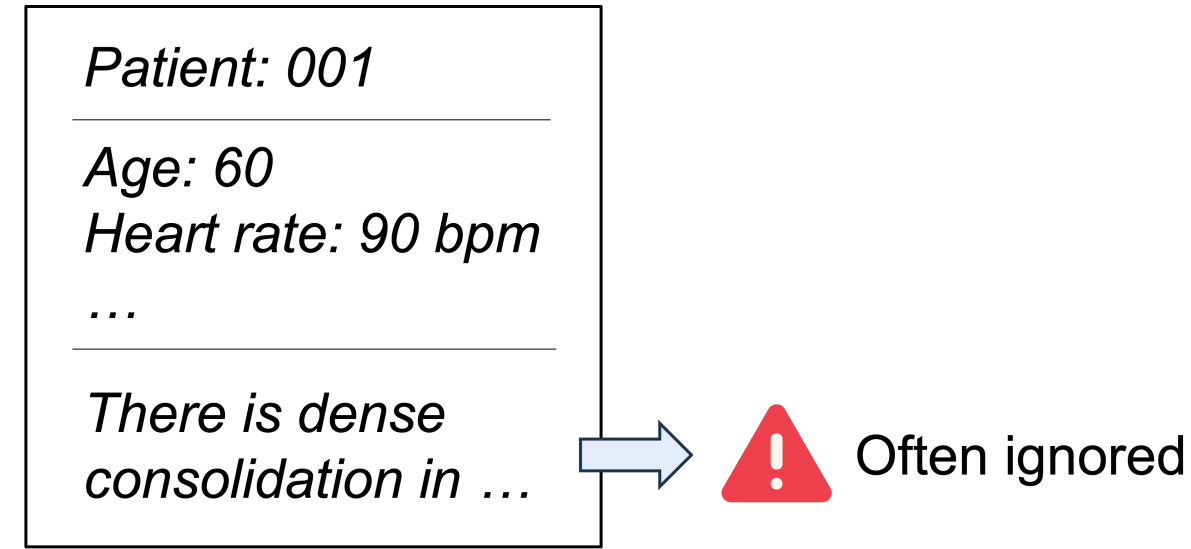


Figure 1. Structured vs. unstructured information in electronic health records (EHRs)

Objectives

To develop a framework that integrates text-derived features from clinical notes with structured electronic health records (EHR) data using **survival analysis** and **Natural Language Processing (NLP)** to improve deterioration prediction.

METHODOLOGY

1. Cohort Construction & Problem Definition

1a. Community-Acquired Pneumonia (CAP) admissions were extracted from the MIMIC-IV ED dataset, as shown in Figure 2a [2].

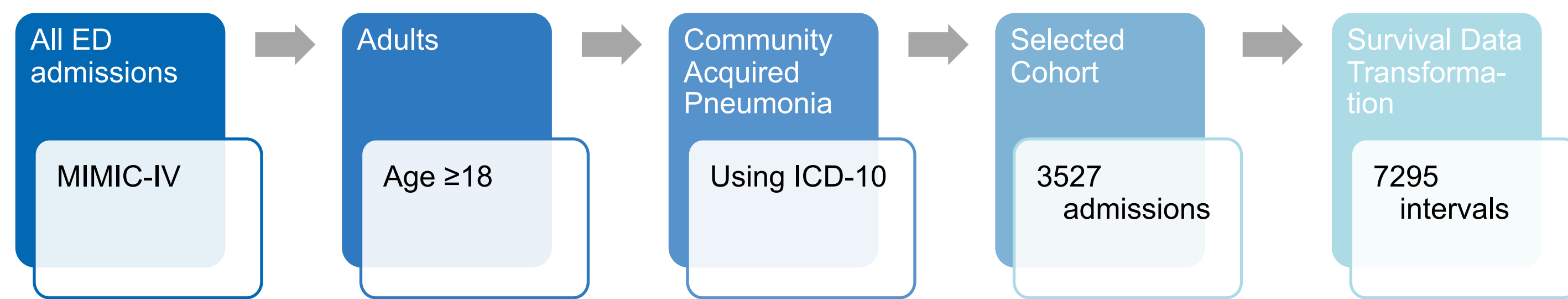


Figure 2. Data preprocessing pipeline from MIMIC-IV ED Dataset

1b. Deterioration within 72h was defined as (i) respiratory failure, (ii) circulatory failure, or (iii) in-hospital death [3].

2. Clinical Text Representation

2a. Radiology reports are encoded using BioBERT to capture clinical semantics [4].

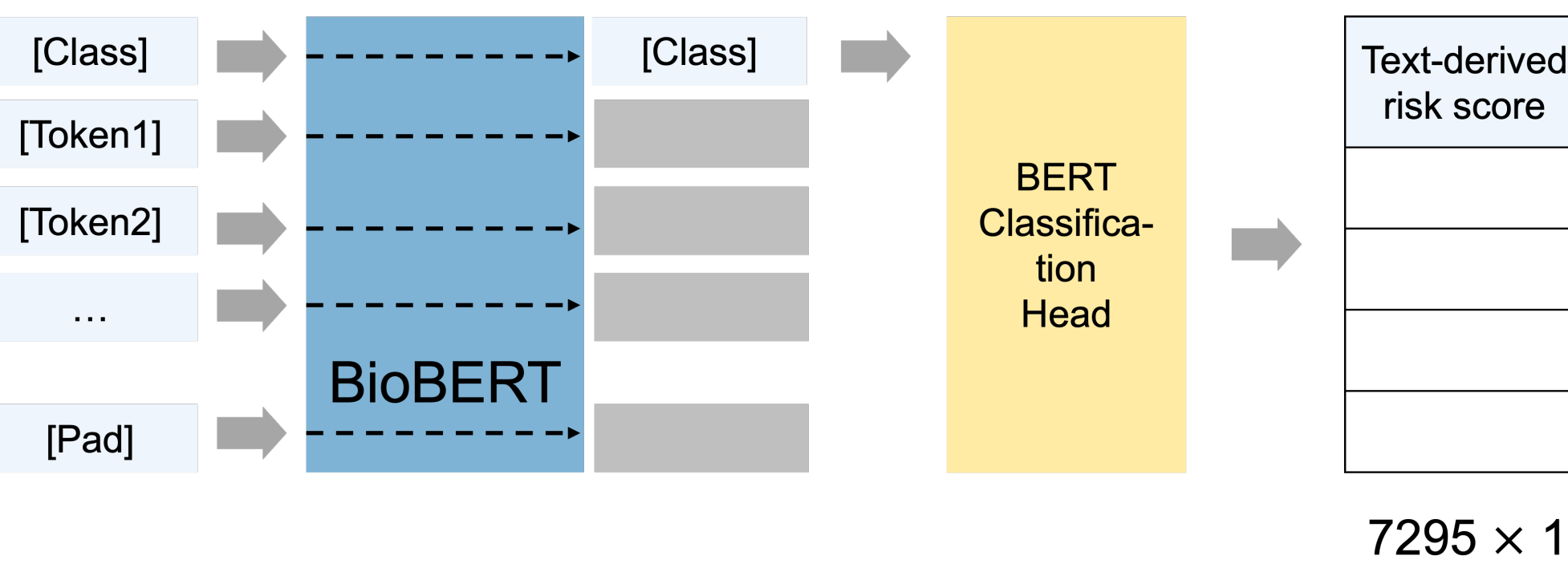


Figure 3. Radiology report encoding using BioBERT

2b. 5-fold GroupKFold cross-validation is used to generate out-of-fold text-derived risk scores, preventing patient-level information leakage.

3. Survival Modeling Framework

Text-derived features are combined with structured EHR data to train survival models for time-to-deterioration prediction.

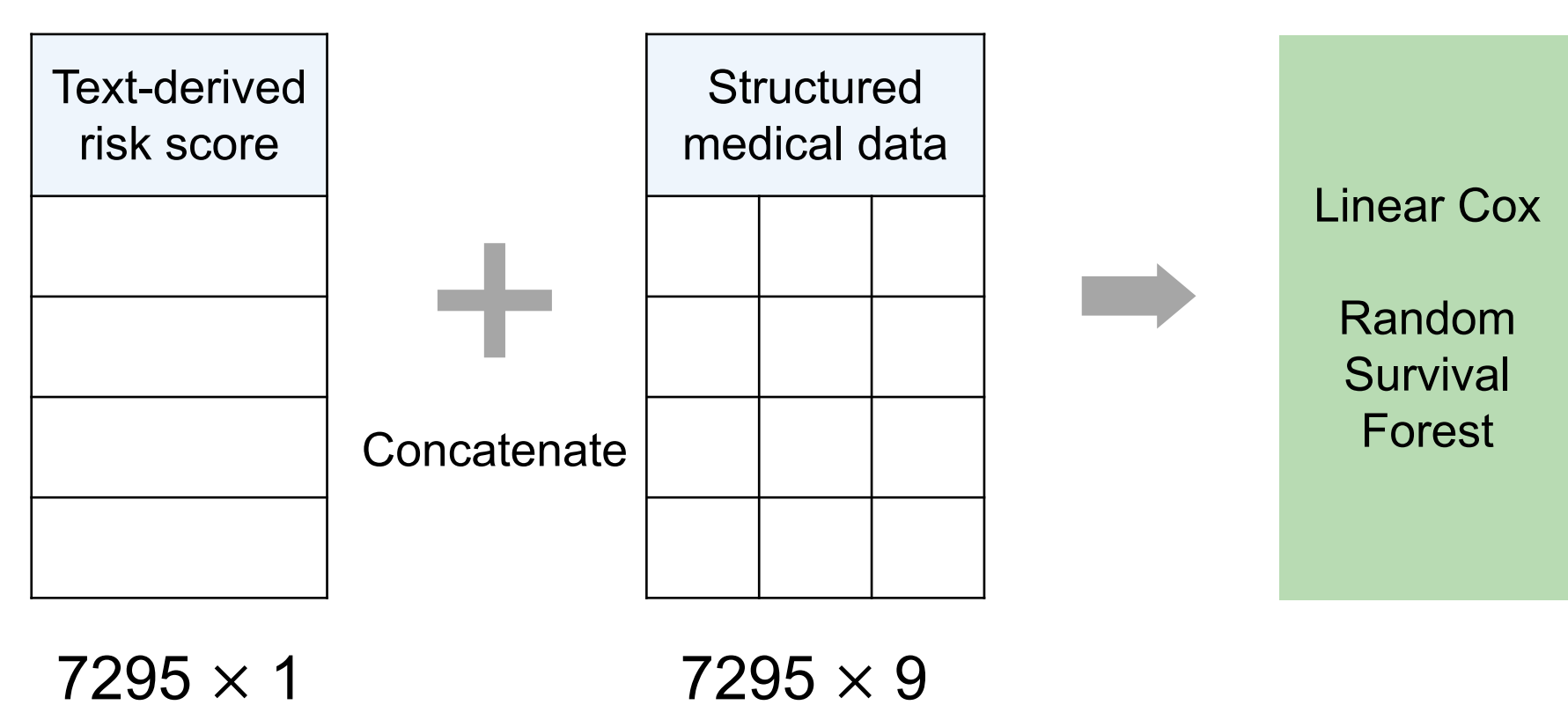


Figure 4. Survival modeling framework combining text and structured data

4. Evaluation

- **Baseline:** Structured EHR model without text-derived features.
- **Models:** Cox (interpretable survival model) and Random Survival Forest (RSF) (non-linear ensemble).
- **Metrics:** Concordance index (C-index) for discrimination and Integrated Brier Score (IBS) for calibration.
- **Data Split:** Admission-level 70/30 split, stratified by event status; performance estimated via bootstrap (N=1000) with 95% CIs.

RESULTS

1. Predictive Performance

Compared to baseline models using structured data only, both Cox and RSF models augmented with text-derived features achieve higher discriminative performance (C-index) and improved calibration (lower IBS).

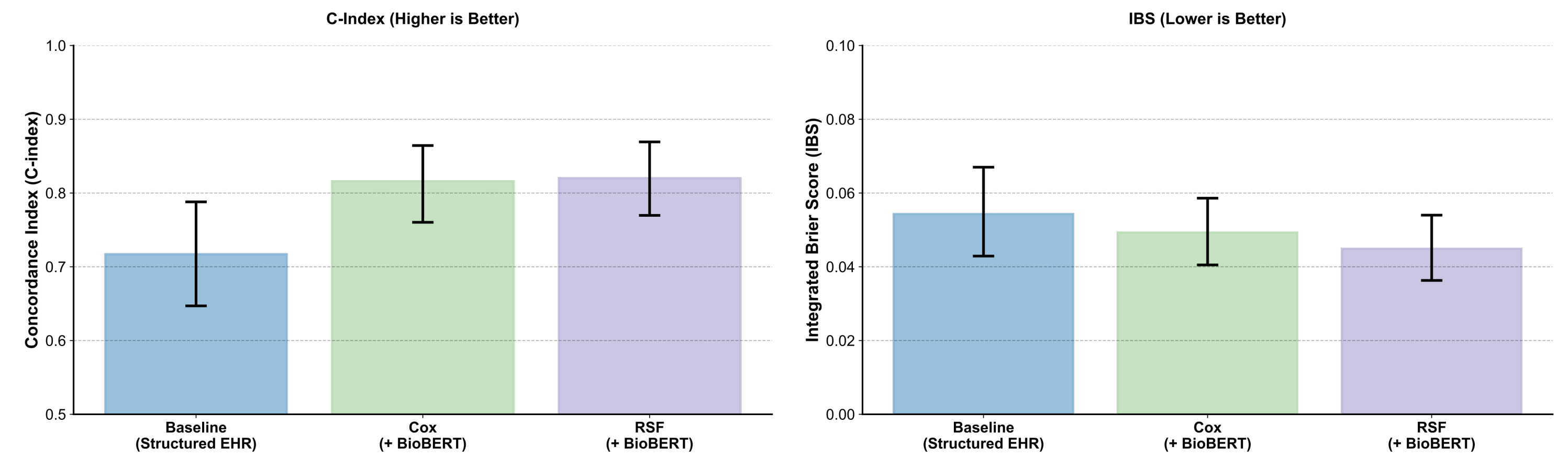


Figure 5. C-index and Integrated Brier Score evaluated from chosen models

Incorporating BioBERT-derived features significantly improves prediction of clinical deterioration compared to structured EHR data alone.

2. Risk Stratification by Text-Derived Score

- High-risk patients exhibit rapid early decline, while low-risk patients maintain high survival probabilities throughout.
- Differences between groups are highly significant (log-rank $p < 0.001$), demonstrating strong clinical stratification.

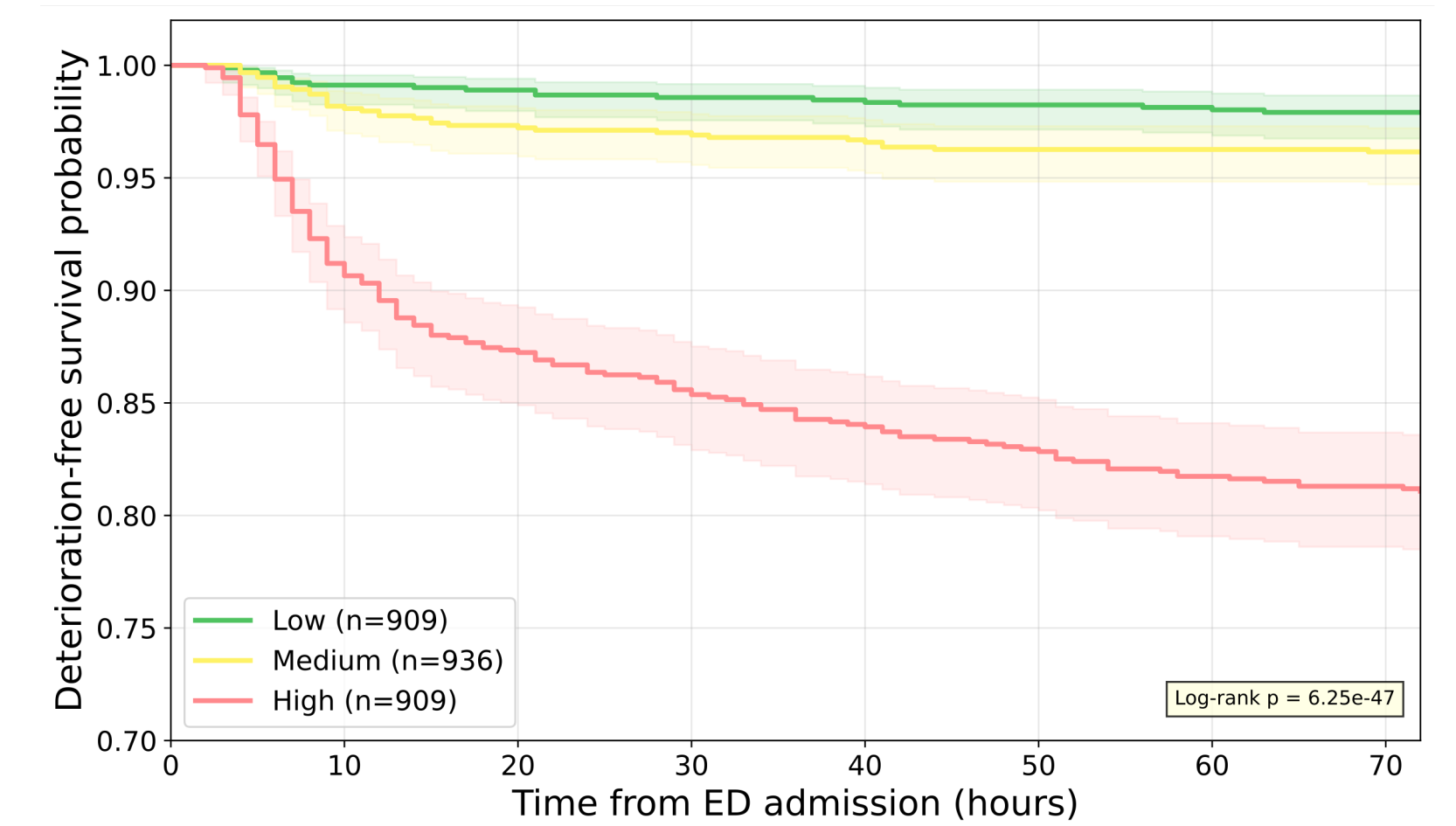


Figure 5. Kaplan-Meier curves by text-derived risk groups

The text-derived risk score effectively stratifies patients into clinically distinct risk groups.

3. Model Interpretability

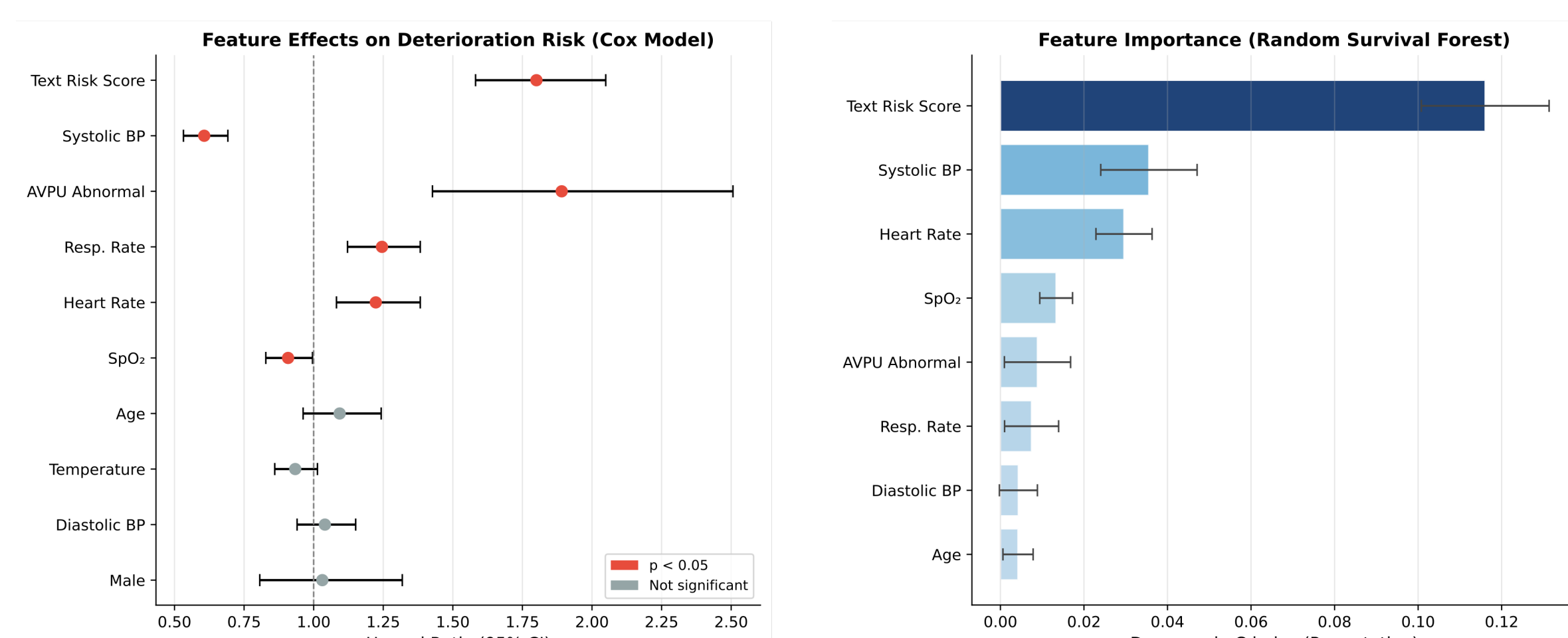


Figure 6. Feature importance from Cox model and Random Survival Forest

Both the Cox model and RSF consistently identify the text-derived risk score as the strongest predictor of deterioration. AVPU abnormal status and key vital signs, such as heart rate and systolic blood pressure, provide additional contribution, while other variables have smaller effects.

Survival models identify clinically meaningful predictors of deterioration risk.

CONCLUSIONS

- Incorporating radiology text via BioBERT significantly enhances prediction of clinical deterioration.
- The model enables robust and clinically meaningful risk stratification over time.
- Text-derived features consistently provide greater predictive value than structured clinical variables.

FUTURE WORK

- To develop multimodal frameworks combining imaging, text, and structured data for improved prediction.
- To conduct external validation to ensure robustness and generalisability across diverse clinical settings.

References

- [1] Royal College of Physicians, "National Early Warning Score (NEWS2): Standardising the assessment of acute-illness severity in the NHS," 2017. [Online]. Available: <https://www.rcp.ac.uk/improving-care/resources/national-early-warning-score-news-2/>
- [2] A. Johnson et al., "MIMIC-IV-ED," PhysioNet, Jan. 2023. [Online]. Available: <https://doi.org/10.13026/5ntk-km72>
- [3] P. Tajernmuang et al., "The National Early Warning Score 2 (NEWS2) to predict early progression to severe community-acquired pneumonia," Tropical Medicine and Infectious Disease, vol. 8, no. 2, p. 68, 2023.
- [4] J. Lee et al., "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, vol. 36, no. 4, pp. 1234–1240, 2020.

